BlastSmart App

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1 Abstract

The mining industry heavily depends on effective blast design to achieve safe, productive ,and cost-effective blast operations. Traditional methods that have been followed by mines over the years such as manual planning , which can be time consuming, often lead to human error. In order to address these limitations or issues, Dwala Global Resources has proposed the development of a Blast-Smart App, a mobile application that enables engineers to perform real-time blast design calculations. This project focuses on the development of two key features of the app, which are the Surface Blasting Design Calculator and the Fragmentation Calculator. The Surface Blasting Design Calculator allows users to calculate blast design parameters such as, burden, stemming length, and actual powder factor by inputting values which are the blasting diameter, average in-hole density, bench height, and spacing to burden ratio. The fragmentation calculator enables the user to estimate the mean fragmentation size based on the blasting parameters provided. The Blast-Smart App aims to improve the accuracy and efficiency of blast design calculations while supporting mining operations through the use of technology .

2 Introdution

Blasting is an essential process in mining as it affects productivity, safety and rock fragmentation. Accurate blast design methods are therefore essential for ensuring safety and cost-effective operations. Manual calculation methods are often too slow and are more likely to result in errors, creating the need for digital technology. This project develops the Blast-Smart App to simplify blast design and fragmentation calculations through a user-friendly mobile app.

3 Background

In South African surface mining, drilling and blasting is the first and most important stage of rock excavation. Effective blast design requires accurate calculation of parameters such as burden, spacing, stemming length, and powder factor to achieve good fragmentation and efficient mining operations.

However, these calculations are often time-consuming, difficult to standardize, and may have human error. To address these challenges, Dwala Global Resources developed the BlastSmart mobile application, which automates blast design calculations. This report presents Phase 1 of the project, focusing on the Surface Blasting Design and Fragmentation Calculator modules.

4 LITERATURE REVIEW

Efficient blast design is fundamental to mining productivity, cost control, and safety, as parameters such as burden, spacing, bench height, stemming length, explosive density, and powder factor directly govern rock fragmentation. Empirical models like the Kuznetsov and Kuz Ram equations offer validated methods for predicting mean fragment size and optimizing blast performance. However, reliance on manual, paper based calculations limits accuracy and speed in the field. By integrating these models into the BlastSmart mobile application, specifically through the Surface Blasting Design and Fragmentation Calculators, blasting engineers can perform rapid, standardized computations in real time. This digitization has the potential to reduce design time, improve fragmentation outcomes, and support better decision making in South African mining operations.

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5 Design

The blast Smart app was design with simple and user-friendly interface that allows user to insert the required blast design parameters. The design follows a simple logical flow where data Is entered and processed using algorithms and calculated results are then displayed. This improves blast design calculation performance. The application consists of two modules, the surface blasting design calculator and fragmentation calculator. Each module is responsible to receiver the user's input values, performing the appropriate calculations and displaying the corresponding output. This improves the overall structure of application and provide flexibility for incorporating blasting tools in future developments.

6 Results

Pseudocodes

6.1. Surface Blasting Design Calculator

FUNCTION SurfaceBlastingDesignCalculator(BlastDiameter, AverageInHoleDensity, SpacingToBurdenRatio, BenchHeight):

- Calculate Stemming length using the required formula
- Calculate Burden using the required formula
- Calculate Actual Powder factor using the required formula

RETURN Stemming length, Burden, Actual Powder Factor

END FUNCTION

6.2. Fragmentation Calculator

FUNCTION FragmentationCalculator(AverageInHoleDensity, BenchHeight, RelativeEffectiveEnergy, StemmingLength):

- Calculate Mean Fragment Size using the required formula

RETURN Mean Fragment Size

END FUNCTION

Results

Table 1: Surface Blasting Design Calculator Results

Input Parameters	Output (Required Blast Design Parameters)
Blasting Diameter (BD)	Burden (B)
Average In-Hole Density (ρ)	Stemming Length (SL)
Spacing-to-Burden Ratio (S/B)	Actual Powder Factor (APF)
Bench Height (BH)	

Input Parameters	Output
Average In-Hole Density (ρ)	Mean Fragment Size (X50)
Bench Height (BH)	
Relative Effective Energy (E)	
Stemming Length (SL)	

Table 2: Fragmentation Calculator Results

7 DISCUSSION OF RESULTS

The developed algorithm demonstrates how the essential blasting parameter can be used in digital applications to automate the blast design parameters. The surface blast design calculator processes the input of the users, including the blast design parameters , to calculate the stemming length, burden and actual powder factor. This blast parameters plays an important role in determining the rock fragmentation and blast efficiency. The fragmentation calculator predicts the mean fragment size by analysing the explosive properties together with blast geometry. This enables the engineers to evaluate the expected fragmentation performance and check whether the rock blasted is suitable for loading, hauling and crushing. Achieving appropriate fragmentation can reduce the need for secondary blasting and decrease energy consumption.

In overall the blast design app will provide a reliable approach to blast design by reducing the need to do manual calculations and limiting human error. As a result, this application has a potential in improving blasting efficiency, operational safety and contribute to a more productive and cost-effective mining operations.

8 CONCLUSION

The BlastSmart application provides a practical and efficient solution for performing blast design calculations. By automating calculations and reducing manual work, it accuracy, saves time and support better blast planning.

9 RECOMMENDATIONS

It is recommended that the BlastSmart application be further developed to incorporate additional modules, specifically Underground Blast Design, Prediction Calculators, and Rapid Calculators, in order to deliver a more comprehensive solution for blasting engineers. Subsequent iterations of the application should implement input validation mechanisms to mitigate erroneous data entry and enhance the precision of computational outputs. Furthermore, the application should undergo evaluation by mining professionals within operational environments to assess its performance, reliability, and usability. Lastly, to ensure sustained accuracy, user-friendliness, and compatibility with evolving technological advancements in the mining sector, regular updates and systematic maintenance are advised.

10 REFERENCES

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